DASE7140 Object Tracking -- Technical Highlights

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Single-pedestrian tracker for sample.mp4 | YOLOv8n + per-frame detection + multi-cue scoring + NCC verification

1. Problem & Challenge

The task is to track one specific pedestrian in sample.mp4 (1920x1080, 25 fps, 403 frames). The scene is crowded with many pedestrians, and the target undergoes severe scale changes (walks away from the camera, shrinking from ~50 kpx to ~5 kpx) and partial occlusion. Hard-coding the target identity is allowed, but the tracker must not drift to other people.

2. Approach Overview

We adopt a per-frame detection paradigm rather than relying on tracker-assigned IDs. Each frame is processed independently through YOLOv8n, and the best-matching person box is selected by a composite scoring function. A final NCC (normalized cross-correlation) template-verification gate rejects false positives that survive the scoring stage.

3. Key Design Decisions

- Auto target selection (frame 0): a heuristic score combining center distance to image bottom-center, bounding-box area, and vertical position. No manual annotation is needed.
- Motion prediction: linear velocity extrapolation from the last 5 tracked frames. During lost states the prediction horizon grows, allowing brief re-acquisition.
- Color constancy: HSV 2-D histogram (16x16 H-S bins, Bhattacharyya distance). The reference histogram is updated with an exponential moving average (alpha = 0.1) to adapt to slow illumination changes without forgetting the original appearance.
- NCC template verification: a multi-scale template pyramid (1.0x, 0.5x, 0.25x) is built from the frame-0 crop. At every candidate box a 1.5x search-window NCC is run. The match is accepted only if the maximum correlation coefficient >= 0.55.
- Dynamic area prior: when the target is lost, expected_area decays by 5% per frame. This prevents the hard area filter from permanently rejecting the correct box after scale change.

4. Iteration & Ablation

Early versions used only HSV color matching and drifted to wrong pedestrians around frame 200 because other people wore similar colours. Adding the NCC firewall solved this: the tracker remains correct at frame 200. After frame 250 it honestly reports "Lost" rather than drifting, because the single static template can no longer match the target under large pose and illumination changes.

5. Failure Analysis

The main failure mode is template staleness. The template is extracted once at frame 0 and never updated. After 250+ frames the target's appearance (pose, lighting angle, scale) has changed so much that NCC drops below 0.55. Attempted fixes -- multi-template libraries (drift pollution) and gradient NCC (insufficient texture) -- were abandoned because they introduced new failure modes worse than honest loss.

6. Model & Resource Choices

- Detector: YOLOv8n (6.3 MB), well under the 10 MB limit.
- No GPU assumption: CUDA_VISIBLE_DEVICES has been removed; PyTorch auto-fallback to CPU works seamlessly.
- Runtime: ~30 s for 403 frames on CPU (Intel i7-class).